# OrthoNu: Revolutionizing Oral Self-Care

**Dr. Sima Yakoby Epstein, Francesca Kirby, Emily Gaw, and Dr. Scott S. De Rossi, OrthoNu**

## Executive Summary

Orthodontic care is advancing rapidly in precision, digital design, and patient experience—but one essential domain remains overlooked: **orthodontic self-care**. Between office visits, millions of orthodontic patients struggle daily with oral pain, irritation, and hygiene challenges that traditional products—wax, rinses, or silicone chews—fail to solve effectively.

The **OrthoNu Oral Relief Collection** (also known as the *Oral Relief Kitz™*) redefines this neglected space. Developed *by orthodontists for orthodontic patients*, it combines scientifically designed, dissolvable, and biocompatible materials that soothe discomfort, protect oral tissues, and promote healing—while improving patient adherence and reducing costly emergency visits.

The Collection's four flagship products—**Chillin' Strips™, Comfort Tape™, mouth-aid™, and OrthoChewz™**—comprise the first clinically inspired, comprehensive self-care system for orthodontic patients. When paired with OrthoNu's **Tweakz®** professional-grade toolkit, these products deliver a 360° solution that elevates patient experience, improves clinical efficiency, and advances OrthoNu's mission to make **mouthcare = healthcare**.

## The Unmet Need: Orthodontic Pain, Emergencies, and Patient Experience

### The Problem

Every orthodontic office faces the same recurring challenge: patients in discomfort, calling for emergency visits due to broken wires, sore gums, or aligner irritation. Nationally, these issues cost practices **over $100,000 annually** in lost productivity and chair time.

Patients—particularly adolescents and their parents—often feel unprepared to manage oral discomfort at home. Without proper self-care tools, they rely on makeshift fixes (e.g., nail clippers for protruding wires, silicone wax for irritation) that are ineffective or unsafe. The result: frustration, interrupted treatment progress, and a negative perception of orthodontic care.

### The Gap in the Market

Traditional dental self-care focuses on **teeth, gums, and breath**—but neglects the unique challenges of **orthodontic self-care**. Braces, aligners, and appliances create new surfaces, friction points, and vulnerabilities that require targeted, safe, and effective management solutions.

Until now, there has been **no cohesive product line** designed specifically for orthodontic patients to manage comfort, healing, and hygiene between visits.

## The OrthoNu Vision: Transforming Orthodontic Self-Care

OrthoNu is not merely creating products—it is **defining a new category** in orthodontics. Founded by **Dr. Sima Yakoby Epstein, DMD**, an orthodontist with more than a decade of clinical experience, the company's mission is to **empower patients** with science-driven, accessible solutions that reduce emergencies, improve outcomes, and elevate the overall standard of care.

### Mission and Standard

- **Vision:** Transform society's understanding that *mouthcare is healthcare*.
- **Mission:** Empower orthodontic patients with dynamic, science-driven care essentials.
- **Standard:** Patient-, science-, and research-driven solutions to manage discomfort and prevent unnecessary emergency visits.

The **OrthoNu Standard** integrates clinical science with consumer design. Each product in the Oral Relief Collection is designed to *function like a medical device but feel like self-care*—safe, effective, and intuitive for at-home use.

## The Oral Relief Collection: Intelligent Comfort for Every Orthodontic Need

The **OrthoNu Oral Relief Kitz™** represents the culmination of years of clinical insight, R&D, and patient feedback. Each product targets a distinct aspect of orthodontic discomfort and healing while collectively forming a comprehensive solution for everyday challenges.

| Product | Tagline / Analogy | Primary Function | Duration | Clinical Use Case |
|---|---|---|---|---|
| **Chillin' Strips™** | "Replaces Silicone Wax" | Soothes irritation & sensitivity to oral mucosa, soft and hard tissues in the oral cavity. | Up to 2 hours | Braces friction, aligner irritation, extraction sites |
| **Comfort Tape™** | "Band-Aid for Teeth & Gums" | Eases gum and tooth discomfort | 2+ hours | Sensitive teeth, gingival inflammation, appliance friction |
| **mouth-aid™** | "Aid for Sores & Ulcers" | Relieves canker sores, ulcers, and mucosal irritation | 2+ hours | Aphthous ulcers, trauma from braces or aligners |
| **OrthoChewz™** | "Replaces Silicone Chewies" | Dissolvable chew to relieve dry mouth and aligner discomfort | Dissolves fully | Seating aligners, salivary stimulation, oral dryness |

**Chillin' Strips™** and **OrthoChewz™** dissolves comfortably in the mouth, releasing soothing and protective ingredients that *treat the cause, not just the symptom*—a significant advancement over traditional silicone waxes or gels.

## The Science Behind OrthoNu's Innovation

### Dissolvable Bioadhesive Matrix

OrthoNu's films and tapes utilize proprietary **dissolvable polymer technology** engineered to adhere gently to enamel or mucosa, maintaining position and moisture balance while delivering comfort compounds for hours. Unlike wax, they *do not fall off, flake, or interfere with speech.*

### Natural and Functional Ingredients

Each formula integrates **biocompatible and naturally derived components** chosen for their proven efficacy in reducing inflammation, sensitivity, and bacterial irritation. Many include **cooling agents and botanical extracts** that soothe soft tissue and provide a clean, refreshing experience.

### Designed by Orthodontists, Validated by Patients

The Oral Relief products were **co-designed with practicing orthodontists** who identified specific pain points in braces and aligner care. The result is a clinically intelligent system that bridges the gap between **oral health** and **orthodontic comfort**—supported by strong patient testimonials and practice integration.

## Clinical and Practice Impact

### Reducing Emergency Visits

By providing patients with safe, effective self-care tools, orthodontic practices can significantly reduce:

- After-hours emergency calls
- In-office visits for wire pokes or soft-tissue injuries
- Unnecessary chair time and workflow disruption

A practice with 200 annual starts can save **over $100,000** in emergency-related costs and regain over **100 clinical hours per year**.

### Enhancing the Patient Experience

Improved comfort correlates directly with:

- Higher patient satisfaction and referrals
- Stronger treatment adherence
- Fewer interruptions in appliance wear
- Reduced anxiety and increased trust in orthodontic providers

### Advancing the Standard of Care

As one orthodontist noted, *"The OrthoNu starter kit should be the standard of care in every practice."* This sentiment is echoed across clinicians and industry leaders—from **Dental Monitoring** to **Colgate-Palmolive**—all recognizing OrthoNu as a pivotal force in redefining patient experience and clinical outcomes.

## Product Highlights

### Chillin' Strips™

The first true alternative to silicone wax. Designed to soothe irritated gums and sensitive oral mucosa, these strips dissolve over two hours, forming a protective, cooling layer that reduces pain and inflammation. Safe for use under mouthguards or aligners and ideal for wounds, extraction sites, or surgical recovery.

### Comfort Tape™

A "bandage for teeth and gums," Comfort Tape delivers long-lasting relief from appliance-induced sensitivity or inflammation. Its natural ingredients calm irritated tissues and create a micro-barrier against friction and bacterial infiltration.

**mouth-aid™**

Formulated to alleviate oral ulcers, canker sores, and mucosal irritation, mouth-aid™ supports tissue healing and comfort. Its gentle formulation can be applied as often as needed—ideal for patients in high-discomfort phases of orthodontic adjustment or oral mucosal compromise.

**OrthoChewz™**

The industry's first dissolvable therapeutic chew. Beyond replacing silicone chewies, OrthoChewz delivers active oral care ingredients that hydrate, freshen, and promote circulation—enhancing aligner seating and comfort while eliminating hygiene concerns associated with silicone.

## The OrthoNu Treatment Starter Kit: Complete Orthodontic Relief

The Oral Relief Collection integrates seamlessly into the **OrthoNu Treatment Starter Kit**—a curated suite of orthodontic care essentials that includes one **Tweakz® toolkit** (for braces or aligners) and the four Oral Relief products.

This full collection empowers patients to **self-manage common orthodontic issues** safely, confidently, and hygienically between visits. Practices report improved satisfaction, better compliance, and fewer after-hours disruptions.

## Regulatory and Manufacturing Integrity

All OrthoNu products are designed under **FDA Class I (510k exempt)** frameworks, ensuring regulatory compliance while maintaining rapid time-to-market. Manufacturing adheres to **ISO 13485** standards, and product testing includes:

- Biocompatibility and safety validation
- Dissolution and stability testing
- Allergenicity and taste profile optimization

This ensures each Oral Relief product meets the **highest clinical and consumer safety benchmarks**.

## Market Position and Growth Potential

While the orthodontic market continues to grow—projected at **~$11 billion globally by 2032**—most products focus on clinical treatment rather than patient experience. OrthoNu uniquely bridges that gap by creating an entirely new **Orthodontic Self-Care** category within the broader oral health market.

## Strategic Outlook

The Oral Relief Collection lays the foundation for OrthoNu's broader oral wellness platform, which integrates:

- **Smart oral technology (NuSmart™)**
- **Microbiome and peptide-based healing collections**
- **Future Class II and III devices for wound management and oral lesion therapy**

Each phase of innovation builds upon the same OrthoNu principle: **scientific empathy**—where cutting-edge design meets real patient needs.

## Social and Clinical Impact

Beyond market impact, the Oral Relief Collection advances OrthoNu's core purpose—to enhance comfort, safety, and empowerment for patients navigating orthodontic treatment. By redefining self-care as an essential extension of clinical care, OrthoNu promotes a more connected, compassionate, and effective model for oral wellness.

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